



**DEPARTMENT OF COMPUTER SCIENCE (BS IT PROGRAM)**  
**COMPUTER GRAPHICS LAB**  
**PROJECT PROPOSAL FORM**

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| 6th                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring 2026                         | BS (IT)             | Lab Engineer:<br>Sir M. Usama |
| <b>TITLE OF PROJECT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                     |                     |                               |
| <b>INTERACTIVE SOLAR SYSTEM SIMULATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                     |                     |                               |
| <b>Group Members</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                     |                     |                               |
| S#                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Student(s) Name                     | Registration Number |                               |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | FAIZA ALTAF                         | 02-235232-036       |                               |
| email:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 02-235232-036@student.bahria.edu.pk | No:                 | -                             |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | HADIQA MEHMOOD                      | 02-235232-007       |                               |
| email:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 02-235232-007@student.bahria.edu.pk | No:                 | -                             |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MAHNOOR SHARIF                      | 02-235232-001       |                               |
| email:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 02-235232-001@student.bahria.edu.pk | No:                 | -                             |
| <b>EXECUTIVE SUMMARY OF PROJECT PROPOSAL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                     |                     |                               |
| <p>Computer Graphics allows interactive visualization of complex systems like the solar system, where multiple bodies move in coordinated orbits, something difficult to understand using static images.</p> <p>This project develops an Interactive 3D Solar System Simulation using OpenGL and GLUT in C++, modeling the Sun, planets, and moons through hierarchical transformations, lighting, and texture mapping. It includes smooth animation, interactive camera controls, and keyboard/mouse input, focusing on visual realism and demonstrating key Computer Graphics concepts.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     |                     |                               |
| <b>PROJECT PURPOSE, SCOPE AND OBJECTIVES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                     |                     |                               |
| <p><b><u>Purpose</u></b><br/>The purpose of this project is to implement and demonstrate fundamental and advanced Computer Graphics concepts through an interactive and visually engaging solar system simulation.</p> <p><b><u>Scope</u></b><br/><b>In-Scope Features:</b></p> <ul style="list-style-type: none"><li>Sun as central object and light source</li><li>Planets revolving around the Sun</li><li>Moons revolving around planets (hierarchical modeling)</li><li>Smooth animation of rotation and revolution</li><li>Lighting effects (ambient, diffuse, specular)</li><li>Day/Night transitions using light intensity</li><li>Texture mapping on planets (minimum 2–3)</li><li>Interactive camera (orbit, zoom, flyby)</li><li>Keyboard and mouse interaction</li><li>Smooth rendering using double buffering</li></ul> <p><b>Out-of-Scope Features:</b></p> <ul style="list-style-type: none"><li>Real-world astronomical accuracy</li><li>Complex physics simulations (gravity calculations)</li><li>Collision detection</li><li>External engines (Unity, Unreal, etc.)</li></ul> <p><b><u>Objectives</u></b></p> <ul style="list-style-type: none"><li>To implement hierarchical modeling for planet-moon systems</li><li>To apply 3D transformations (translation, rotation, scaling)</li><li>To demonstrate lighting and shading techniques</li><li>To implement texture mapping for realistic surfaces</li><li>To develop interactive camera controls</li><li>To ensure smooth animation using timer-based updates</li><li>To create a visually appealing and user-friendly simulation</li></ul> |                                     |                     |                               |



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**PROJECT DESCRIPTION (most important part)**

The project is a **3D interactive simulation of the solar system** that demonstrates multiple Computer Graphics concepts through a structured and modular implementation.

**System Features**

**1. Solar System Structure**

The system will consist of the Sun at the center with multiple planets orbiting around it. Some planets will also have moons orbiting them, demonstrating hierarchical relationships.

**2. Orbital Motion & Animation**

Planetary motion will be implemented using transformation functions:

- Rotation for spinning planets
- Translation for orbiting
- Smooth animation

Users can control animation speed dynamically.

**3. Hierarchical Modeling**

Planets and moons will be modeled using parent-child relationships:

- Sun → Planet → Moon

This will be implemented using OpenGL matrix stack operations.

**4. Lighting & Shading**

- Sun will act as a primary light source
- Additional ambient light will be used
- Lighting components:
  - Ambient
  - Diffuse
  - Specular
- Smooth day/night transitions will be simulated by adjusting light intensity

**5. Texture Mapping**

- Planets such as Earth and Mars will have textures applied
- BMP images will be used for mapping
- Enhances realism of the simulation

**6. Camera System**

- Orbit camera around solar system
- Zoom in/out functionality
- Flyby navigation through scene
- Controlled via keyboard and mouse

**7. User Interaction (Clean UI)**

**Keyboard Controls:**

- Increase/decrease speed
- Pause/resume animation

**Mouse Controls:**

- Rotate camera
- Adjust viewing angle

System will ensure smooth and responsive interaction without lag.

**8. Scene Enhancements**

- Starfield background using points
- Smooth rendering using double buffering
- Balanced scaling of objects for visual clarity

**Data Processing / Logic**

- Transformation matrices for motion
- Timer-based animation updates



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- Input handling via callbacks
- Texture loading and application

**Justification (Why This Project Should Be Accepted)**

- Covers all mandatory requirements from the course
- Demonstrates multiple advanced CG concepts
- Highly interactive and visually appealing
- Uses clean modular implementation
- Includes bonus-level enhancements (smooth animation, lighting transitions, UI quality)

**Course Concepts to be Used:**

- OpenGL primitives (spheres, points)
- 3D transformations (translate, rotate, scale)
- Hierarchical modeling (matrix stack)
- Lighting models (ambient, diffuse, specular)
- Multiple light sources
- Texture mapping
- Camera and projection systems
- Animation using timers
- Keyboard and mouse interaction
- Double buffering

**TEAM PROFILE**

**Team Members:**

- Faiza Altaf
- Hadiqa Mehmood
- Mahnoor Sharif

All team members will work collaboratively on every aspect of the project including programming, design, testing, presentation, and documentation.

The team believes in equal contribution, effective communication, and teamwork to successfully complete the project. Each member will support one another to ensure high-quality output and timely completion of all project tasks.

**ASSUMPTIONS AND CONSTRAINTS**

**Assumptions:**

- OpenGL and GLUT libraries are available on lab systems
- All team members have basic knowledge of C++
- Required textures and resources will be accessible

**Constraints:**

- Must strictly use OpenGL + GLUT only
- Limited development time
- Performance limitations of lab machines
- No use of external graphics engines
- Project must compile and run without errors



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| <b>PROJECT DELIVERABLES <span style="color: red;">NOT CHANGEABLE</span></b>                                                                                                                                                                                          |
| <b>Deliverables include</b> <ul style="list-style-type: none"><li>● Software Project Proposal.</li><li>● Project progress</li><li>● Project report</li><li>● Team member's work, as per their contribution, you should have to be honest with your future.</li></ul> |
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| <b>For Teacher Use Only</b> |
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| <b>REMARKS</b> |
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|--------------------------------|--|------------------|--|-------------|--|
| <b>Course Teacher<br/>Name</b> |  | <b>Signature</b> |  | <b>Date</b> |  |
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